

Pure Mathematics (40 marks)

- 1** When a solid turns into a gas without first becoming a liquid, the process is called sublimation. As a spherical mothball sublimates, its volume, in cm^3 , decreases at a rate, in cm^3 per day, proportional to its surface area.

Show that the radius of the mothball decreases at a constant rate. [3]

[It is given that a sphere of radius r has surface area $4\pi r^2$ and volume $\frac{4}{3}\pi r^3$.]

1 [3]	Let V and A be the volume and surface area of the spherical mothball respectively. $\frac{dV}{dt} = kA, k < 0 \quad \dots\dots\dots (1)$ $V = \frac{4}{3}\pi r^3$ $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} = A \frac{dr}{dt} \quad \dots\dots\dots (2)$ $(1) = (2) \Rightarrow \frac{dr}{dt} = k < 0$ $\therefore$ The radius decreases at a constant rate (shown).	Indicate $k < 0$.
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- 2** The curve C has equation

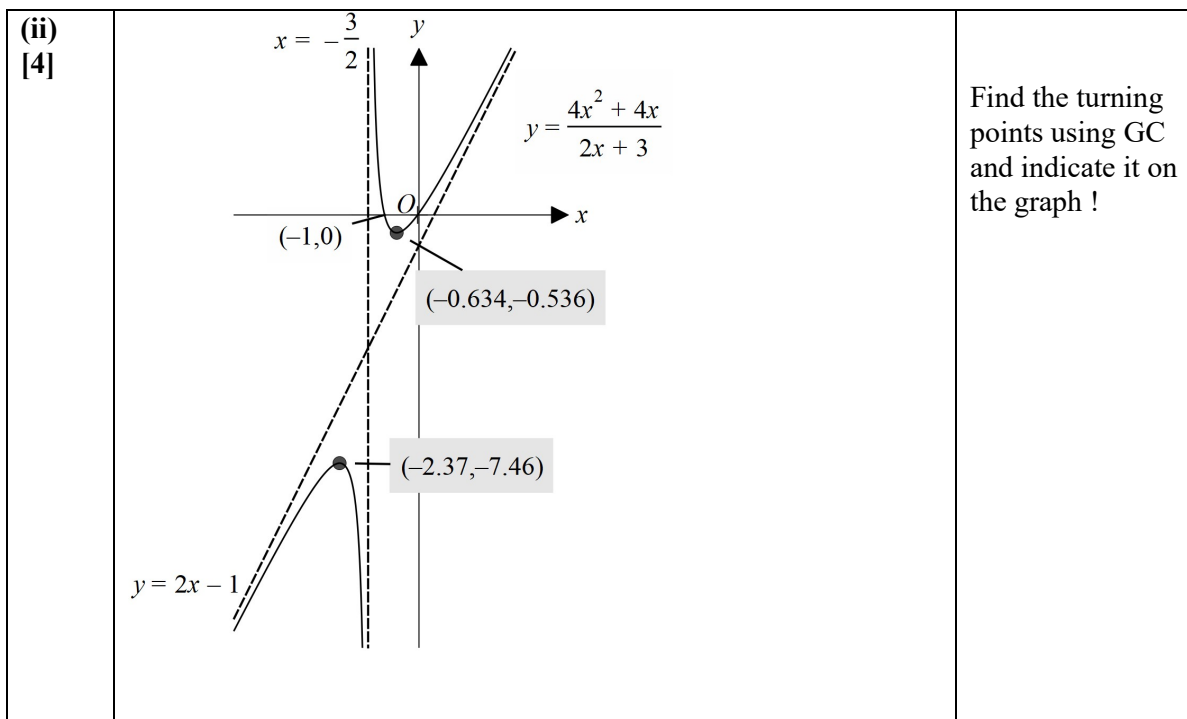
$$y = \frac{(ax+1)(4x+b)}{2x+3}, \quad x \in \mathbb{R}, x \neq -\frac{3}{2},$$

where a and b are constants. It is given that $y = 2x - 1$ is an asymptote of C .

- (i)** Find the values of a and b . [3]

- (ii)** Sketch C , labelling all relevant point(s) and stating the equations of any asymptotes. [4]

2(i) [3]	$y = \frac{(ax+1)(4x+b)}{2x+3} = (2x-1) + \frac{c}{2x+3}$ $\therefore \frac{(ax+1)(4x+b)}{2x+3} = \frac{(2x-1)(2x+3)+c}{2x+3}$ $(ax+1)(4x+b) = (2x-1)(2x+3)+c$ By comparing coefficients of $x^2 : a = 1,$ $x : ab + 4 = 6 - 2 \Rightarrow ab = 0 \Rightarrow b = 0$	Long division is a longer method.
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- 3 The position vectors of the points U and V with respect to the origin O are $\mathbf{u}$ and $\mathbf{v}$ respectively, where $\mathbf{u}$ and $\mathbf{v}$ are non-zero and non-parallel vectors.

- (i) Show that $(\mathbf{u} + \mathbf{v}) \times (\mathbf{v} - 2\mathbf{u})$ can be written as $k(\mathbf{u} \times \mathbf{v})$, where k is a constant to be determined, justifying your working. [2]

The vectors $\mathbf{u}$ and $\mathbf{v}$ are now given by

$$\mathbf{u} = \mathbf{i} - \mathbf{j} + 2\mathbf{k} \text{ and } \mathbf{v} = a\mathbf{i} + b\mathbf{j},$$

where a and b are constants.

- (ii) If $a = 2$ and $b = 3$, find a unit vector parallel to $(\mathbf{u} + \mathbf{v}) \times (\mathbf{v} - 2\mathbf{u})$. [2]
- (iii) If $\mathbf{v}$ is perpendicular to the vector $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, find $\mathbf{v}$ in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and b . Interpret the solution geometrically. [3]

3(i) [2]	$ \begin{aligned} (\mathbf{u} + \mathbf{v}) \times (\mathbf{v} - 2\mathbf{u}) &= (\mathbf{u} \times \mathbf{v}) - 2(\mathbf{u} \times \mathbf{u}) + (\mathbf{v} \times \mathbf{v}) - 2(\mathbf{v} \times \mathbf{u}) \\ &= (\mathbf{u} \times \mathbf{v}) + 2(\mathbf{u} \times \mathbf{v}) \\ &\quad (\because \mathbf{u} \times \mathbf{u} = \mathbf{0}, \mathbf{v} \times \mathbf{v} = \mathbf{0}, \mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})) \\ &= 3(\mathbf{u} \times \mathbf{v}) \quad \therefore k = 3 \end{aligned} $	This is a “show” question so explicit working/ reasoning should be shown. Important: $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$
(ii) [2]	$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} -6 \\ 4 \\ 5 \end{pmatrix}$ A unit vector parallel to $(\mathbf{u} + \mathbf{v}) \times (\mathbf{v} - 2\mathbf{u})$ is $\pm \frac{1}{\sqrt{77}} \begin{pmatrix} -6 \\ 4 \\ 5 \end{pmatrix}$.	Students must represent vectors appropriately. The “tilde” ~ underneath the letter should not be forgotten.

(iii) [3]	Since $\mathbf{v}$ is perpendicular to the vector $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$, $\begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = 0$ $a - 2b = 0$ $a = 2b$ $\mathbf{v} = \begin{pmatrix} 2b \\ b \\ 0 \end{pmatrix} = b \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ The solution represents the position vector of any point on the line which passes through the origin and is parallel to the vector $2\mathbf{i} + \mathbf{j}$.	Explain/describe properly what $\mathbf{v}$ represents.
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4 A curve C has parametric equations

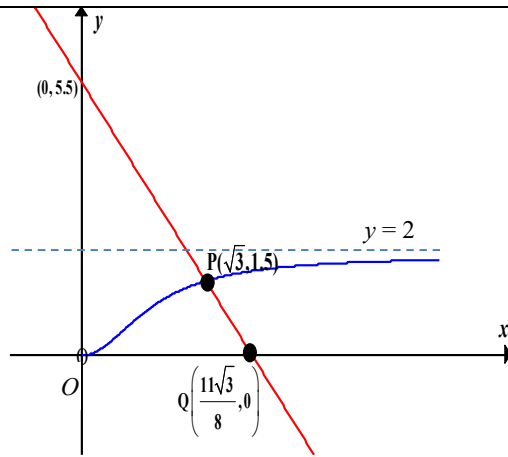
$$x = \cot \theta, \quad y = 2 \cos^2 \theta,$$

where $0 < \theta \leq \frac{1}{2}\pi$.

- (i) The point P on C has parameter $\theta = \frac{1}{6}\pi$. Show that the gradient of the tangent at P is $\frac{\sqrt{3}}{4}$. Hence find the equation of the normal, L , at P . [3]
- (ii) Sketch C and L on the same diagram, stating the coordinates of the points where the line L cuts the curve C and the x -axis respectively. [2]
- ~~(iii) Find the exact area of the region bounded by the curve C , the positive y -axis and the line $y = \frac{3}{2}$. [3]~~
- (iv) Show that the Cartesian equation of C is given by $y = \frac{2x^2}{a+x^b}$, where a and b are constants to be determined.
- (v) The region bounded by the curve C , the line L , the positive x -axis and the line $x = \frac{1}{\sqrt{3}}$ is rotated through 2π radians about the x -axis. Find the volume of the solid formed. [3]

4(i) [3]	$x = \cot \theta \Rightarrow \frac{dx}{d\theta} = -\operatorname{cosec}^2 \theta$ $y = 2 \cos^2 \theta \Rightarrow \frac{dy}{d\theta} = -4 \sin \theta \cos \theta$ $\frac{dy}{dx} = \frac{-4 \sin \theta \cos \theta}{-\operatorname{cosec}^2 \theta} = 4 \sin^3 \theta \cos \theta$ When $\theta = \frac{\pi}{6}$, $\frac{dy}{dx} = 4 \left(\frac{1}{2}\right)^3 \left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{4}$ (shown) At P, $x = \sqrt{3}$ $y = \frac{3}{2}$. Equation of L, the normal at P, is $y - \frac{3}{2} = -\frac{4}{\sqrt{3}}(x - \sqrt{3}) \quad \text{or} \quad y = -\frac{4}{\sqrt{3}}x + \frac{11}{2}$	In a show question, sufficient working must be demonstrated.
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(ii)
[2]



Label the graph clearly with all relevant features.

(iii)
[3]

$$\begin{aligned}
 \text{Area} &= \int_0^{\frac{3}{2}} x \, dy & y=0 &\Rightarrow \cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} \\
 &= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \cot \theta \frac{dy}{d\theta} d\theta & y=\frac{3}{2} &\Rightarrow \cos \theta = \frac{\sqrt{3}}{2} (\because 0 < \theta \leq \frac{\pi}{2}) \Rightarrow \theta = \frac{\pi}{6} \\
 &= \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \frac{\cos \theta}{\sin \theta} (4 \cos \theta (-\sin \theta)) d\theta \\
 &= -4 \int_{\frac{\pi}{2}}^{\frac{\pi}{6}} \cos^2 \theta d\theta \\
 &= 4 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1 + \cos 2\theta}{2} d\theta \\
 &= 2 \left[\theta + \frac{\sin 2\theta}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \\
 &= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}
 \end{aligned}$$

Alternatively, use Cartesian form of the equation (found in (iv))

$$\begin{aligned}
 \text{Area} &= (\sqrt{3})(1.5) - \int_0^{\sqrt{3}} \frac{2x^2}{1+x^2} dx \\
 &= \frac{3\sqrt{3}}{2} - \int_0^{\sqrt{3}} \frac{2(1+x^2-1)}{1+x^2} dx \\
 &= \frac{3\sqrt{3}}{2} - \int_0^{\sqrt{3}} 2 - \frac{2}{1+x^2} dx \\
 &= \frac{3\sqrt{3}}{2} - \left[2x - 2 \tan^{-1} x \right]_0^{\sqrt{3}} \\
 &= \frac{3\sqrt{3}}{2} - \left[2\sqrt{3} - \frac{2\pi}{3} \right] \\
 &= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}
 \end{aligned}$$

(iv) [2]	$x = \cot \theta \Rightarrow x^2 = \cot^2 \theta$ $y = 2 \cos^2 \theta \Rightarrow y = 2 - 2 \sin^2 \theta \Rightarrow \operatorname{cosec}^2 \theta = \frac{2}{2-y}$ $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$ $\frac{2}{2-y} = 1 + x^2$ $2 = 2 + 2x^2 - y - x^2 y$ $y = \frac{2x^2}{1+x^2}, \quad x > 0 \quad (\because 0 < \theta \leq \frac{\pi}{2} \Rightarrow x > 0)$	
(v) [2]	Volume of solid = $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \pi y^2 \, dx + \frac{1}{3} \pi \left(\frac{3}{2}\right)^2 \left(\frac{3\sqrt{3}}{8}\right)$ $= \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \pi \left(\frac{2x^2}{1+x^2}\right)^2 \, dx + \frac{1}{3} \pi \left(\frac{3}{2}\right)^2 \left(\frac{3\sqrt{3}}{8}\right)$ $= 6.17 \text{ units}^3$ Alternatively Volume of solid = $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \pi y^2 \, dx + \frac{1}{3} \pi \left(\frac{3}{2}\right)^2 \left(\frac{3\sqrt{3}}{8}\right)$ $= \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} \pi (2 \cos^2 \theta)^2 (-\operatorname{cosec}^2 \theta \, d\theta) + \frac{1}{3} \pi \left(\frac{3}{2}\right)^2 \left(\frac{3\sqrt{3}}{8}\right)$ $= 6.17 \text{ units}^3$	Read the question slowly/ clearly to identify the correct “region” to find the volume. Little or NO credit can be given if an incorrect region is used.

5 A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_r = \begin{cases} 0 & \text{for } r = 2, \\ f(r-1) - 2f(r-3) & \text{for } r \text{ even, } r \neq 2, \\ f(r) & \text{for } r \text{ odd.} \end{cases}$$

(i) Use the method of differences to find $\sum_{r=1}^{2n} u_r$. [3]

It is given that $f(r) = \ln\left(\frac{r+1}{r}\right)$.

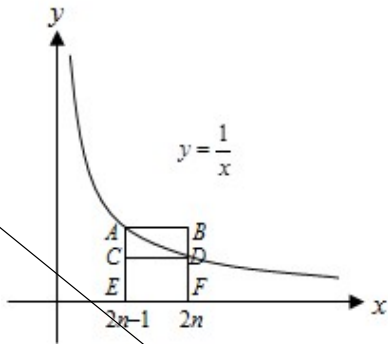
(ii) Use your answer to part (i) to show that $\sum_{r=1}^{2n} u_r = -\ln 2 + 2 \ln\left(1 + \frac{1}{2n-1}\right)$. Hence state the value of the sum to infinity. [2]

(iii) Find the smallest value of n for which $\sum_{r=1}^{2n} u_r$ is within 10^{-2} of the sum to infinity. [2]

(iv) By considering the graph of $y = \frac{1}{x}$ for $x > 0$, show, with the aid of a sketch, that $\frac{1}{2n} < u_{2n-1} < \frac{1}{2n-1}$, $n \in \mathbb{Z}^+$. [3]

(i)		
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[3]	$\sum_{r=1}^{2n} u_r = f(1)$$+0$$+f(3)$$+f(3) - 2f(1)$$+f(5)$$+f(5) - 2f(3)$$+f(7)$$+f(7) - 2f(5)$$+...$$+f(2n-3)$$+f(2n-3) - 2f(2n-5)$$+f(2n-1)$$+f(2n-1) - 2f(2n-3)$$= -f(1) + 2f(2n-1)$	Consider odd and even values. Candidates who are unable to make progress should review piecewise functions.						
(ii) [2]	$\sum_{r=1}^{2n} u_r = -f(1) + 2f(2n-1)$$= -\ln\left(\frac{2}{1}\right) + 2\ln\left(\frac{(2n-1)+1}{2n-1}\right)$$= -\ln 2 + 2\ln\left(1 + \frac{1}{2n-1}\right) \text{ (shown)}$ As $n \rightarrow \infty$, $\frac{1}{2n-1} \rightarrow 0$, $1 + \frac{1}{2n-1} \rightarrow 1$, $\ln\left(1 + \frac{1}{2n-1}\right) \rightarrow 0$, $\sum_{r=1}^{2n} u_r \rightarrow -\ln 2$. Sum to infinity is $-\ln 2$.	While an explanation was not demanded, candidates should note correct reasoning here.						
(iii) [2]	$\left \sum_{r=1}^{2n} u_r - (-\ln 2)\right < 10^{-2}$$2\ln\left(1 + \frac{1}{2n-1}\right) < 0.01 \text{ } (\because 1 + \frac{1}{2n-1} > 1 \Rightarrow \ln\left(1 + \frac{1}{2n-1}\right) > 0)$ From GC, $n > 100.25$ Or alternatively <table><tr><td>n</td><td>$2\ln\left(1 + \frac{1}{2n-1}\right)$</td></tr><tr><td>100</td><td>$0.010025 > 0.01$</td></tr><tr><td>101</td><td>$0.0099256 < 0.01$</td></tr></table> $\therefore$ Smallest value of $n = 101$	n	$2\ln\left(1 + \frac{1}{2n-1}\right)$	100	$0.010025 > 0.01$	101	$0.0099256 < 0.01$	Candidates should note an absolute difference is required.
n	$2\ln\left(1 + \frac{1}{2n-1}\right)$							
100	$0.010025 > 0.01$							
101	$0.0099256 < 0.01$							
(iv) [3]	From the graph of $y = \frac{1}{x}$ for $x > 0$,							

 Area of rectangle CDFE $< \int_{2n-1}^{2n} \frac{1}{x} dx < \text{Area of rectangle ABFE}$ $(1) \left(\frac{1}{2n} \right) < [\ln x]_{2n-1}^{2n} < (1) \left(\frac{1}{2n-1} \right)$ $\frac{1}{2n} < \ln \frac{2n}{2n-1} < \frac{1}{2n-1}$ $\ln \frac{2n}{2n-1} = \ln \frac{(2n-1)+1}{2n-1} = u_{2n-1} \text{ since } 2n-1 \text{ is odd,}$ $\therefore \frac{1}{2n} < u_{2n-1} < \frac{1}{2n-1} \text{ (shown).}$	Compare the integral with rectangle above and below Candidates to note the connection between $y = \frac{1}{x}$ and $y = \ln x$
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Statistics (60 marks)

- 6 For events A and B it is given that $P(A) = 0.5$ and $P(B') = 0.35$.
- (a) Given that events A and B are not independent, find the range of values of $P(A \cup B)$. [2]
- (b) Given that $P(A | B') = 0.6$, find
- (i) $P(A \cup B)$, [3]
- (ii) $P(A' \cup B')$. [1]

(a) [2]	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.5 + 0.65 - P(A \cap B)$ $= 1.15 - P(A \cap B)$ Since $0 \leq P(A \cup B) \leq 1$, $0.15 \leq P(A \cap B) \leq \min\{P(A), P(B)\} \Rightarrow 0.15 \leq P(A \cap B) \leq 0.5$ If A and B are independent, $P(A \cap B) = (0.5)(0.65) = 0.325 \Rightarrow P(A \cup B) = 0.825$ $\therefore 0.65 \leq P(A \cup B) \leq 1$ and $P(A \cup B) \neq 0.825$. Range of $P(A \cup B)$ is $[0.65, 1] \setminus \{0.825\}$.	Students to note that $P(A \cap B) = P(A)P(B)$ is true ONLY when A and B are independent. A and B are NOT independent so $P(A \cup B)$ cannot take value 0.825
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(b)(i) [3]	$P(A' B') = 1 - P(A B')$ $= 1 - 0.6$ $= 0.4$ $P(A' \cap B') = P(B') P(A' B')$ $= (0.35)(0.4)$ $= 0.14$ $P(A \cup B) = 1 - P(A' \cap B')$ $= 1 - 0.14$ $= 0.86$ Alternatively, $P(A \cap B') = P(B') P(A B')$ $= (0.35)(0.6)$ $= 0.21$ $P(A \cup B) = P(B) + P(A \cap B')$ $= (1 - 0.35) + 0.21$ $= 0.86$	Candidates should use the Venn diagrams to analyze the relation of A and B. Candidates should understand the difference between $P(A \cap B)$ and $P(A \cup B)$.
(b)(ii) [1]	$P(A' \cup B') = P(A') + P(B') - P(A' \cap B')$ $= 0.5 + 0.35 - 0.14$ $= 0.71$ Alternatively, $P(A' \cup B') = 1 - P(A \cap B) = 1 - [P(A) - P(A \cap B')]$ $= 1 - [0.5 - 0.21]$ $= 0.71$	Do not confuse De Morgan's Laws, $P(A' \cup B') \neq P(A \cap B)$ $P(A' \cup B') \neq 1 - P(A \cup B).$

- 7 The speed of the surface of a tennis court, affected by a variety of factors, including the physical makeup of the court, can be measured using the Court Pace Index (CPI). The CPI of a court varies with every match played on it. A higher mean CPI for a court indicates a faster court surface.

For the rest of this question, assume that data is collected under similar conditions.

At a tennis club, the CPI of the main court is a continuous random variable X . Based on past readings, the standard deviation of X is k and the expected value of X is 33.8.

- (i) A random sample of 50 readings is to be taken. When $k = 1.5$, estimate the probability that the mean value of X , for this sample, will be at most 0.5 units from 34. [2]

In January 2018, the main court was resurfaced with a new physical makeup.

- (ii) The club management claims that the new physical makeup of the resurfaced main court increases its mean CPI. A random sample of 60 readings were taken from the resurfaced main court after January 2018 and the mean CPI of this sample is found to be 34.3.

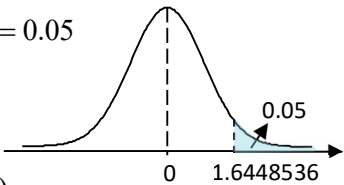
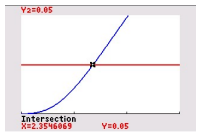
A test is carried out, at the 5% level of significance, to determine whether the resurfaced main court has a faster court surface.

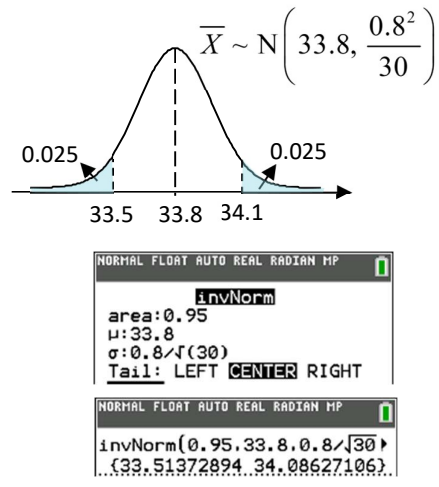
Find the set of possible values of k for which the result of the test should be accepted. You should state your hypotheses clearly. [3]

- (iii) The club management later obtained information that the value of k is 0.8 and that X follows a normal distribution. Another random sample of 30 readings were taken from the resurfaced main court after January 2018 and the mean CPI of this new sample is found to be $\bar{x}$.

A test is carried out, at the 5% level of significance, to determine whether the mean CPI of the resurfaced main court has changed.

Find the set of possible values of $\bar{x}$ for which the result of the test should not be accepted. You should state your hypotheses clearly. [3]

(i) [2]	Since $n=50$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(33.8, \frac{1.5^2}{50}\right)$ approximately. $P(\bar{X} - 34 < 0.5) = P(33.5 < \bar{X} < 34.5) = 0.921$ (3s.f.) Remarks: It is incorrect to claim that X is normally distributed by Central Limit Theorem. Central Limit theorem is used to find the approximate distribution of $\bar{X}$ (i.e. a random sample mean) when the distribution of X is either unknown or not normal. It is not for finding the distribution of X. It holds only when the sample size is large.
(ii) [3]	$H_0 : \mu = 33.8$ $H_1 : \mu > 33.8$ Under H_0, since $n = 60$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(33.8, \frac{k^2}{60}\right)$ approximately At 5% level of significance, to accept test, we reject H_0, $p\text{-value} \leq 0.05$ $P(\bar{X} \geq 34.3) \leq 0.05$ $P(Z \geq \frac{34.3 - 33.8}{k / \sqrt{60}}) \leq 0.05$ From GC, $P(Z \geq 1.6448536) = 0.05$ From the diagram, $\frac{34.3 - 33.8}{k / \sqrt{60}} \geq 1.6448536$ $k \leq 2.3545 = 2.35$ (3s.f.)   From GC, $k \leq 2.35$ (3s.f.) Hence, the set of values of k is $\{k \in \mathbb{R} : 0 < k \leq 2.35\}$. Remarks: - In this hypothesis testing, we are testing on the unknown population mean μ. Thus the null and alternative hypothesis should only involve μ and the value that we are 'suggesting' that it takes. i.e. $H_0 : \mu = 33.8$ $H_1 : \mu > 33.8$ - To do the test, we need to use the distribution of $\bar{X}$ as we are using a sample result (namely the mean of a particular sample = $\bar{x}$) to determine if we are able to reject or not reject the hypothesis at a particular level of significance. As such, the distribution of $\bar{X}$ should only come after the hypothesis is stated.

	• $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$ approximately via Central Limit Theorem or is exactly normal (if X is normal). Also, $E(\bar{X}) = E(X)$ and hence μ_0 should take the value of the population mean μ. However, since the value of μ is unknown, in hypothesis testing, we work under the assumption that μ takes on the value as stated in the null hypothesis and tries to find evidence against it. It is thus important to write the words 'Under H_0' and hence your working should read $H_0: \mu = 33.8$ $H_1: \mu > 33.8$ Under H_0, $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$ where $\mu_0 = 33.8$, ... • Common error - wrong expression for p-value and confusion in algebraic manipulation: p-value refers to the chance of obtaining 'this kind' of sample result (see the notes for proper definition). The sample mean $\bar{x}$ based on an obtained sample was 34.3. Thus $p\text{-value} = P(\bar{X} \geq 34.3)$ (i.e. the chance of sample mean being 34.3 or higher). Thus for (ii), it should read To reject H_0, $p\text{-value} \leq 0.05$ $P(\bar{X} \geq 34.3) \leq 0.05$ $P(Z \geq \frac{34.3 - 33.8}{k / \sqrt{60}}) \leq 0.05$ (standardisation using $\bar{X} \sim N\left(33.8, \frac{k^2}{60}\right)$) $\frac{34.3 - 33.8}{k / \sqrt{60}} \geq 1.6448536$ (note the changes in inequality sign) $k \leq 2.35$ (3s.f.)
(iii) [3]	$H_0: \mu = 33.8$ $H_1: \mu \neq 33.8$ Perform a 2-tail test at 5% level of significance. Under H_0, $\bar{X} \sim N\left(33.8, \frac{0.8^2}{30}\right)$. For the test to be not accepted, we do not reject H_0. Now $P(c_1 < \bar{X} < c_2) = 0.95$, where c_1 and c_2 are symmetric about the mean 33.8 $\therefore$ The set of values of $\bar{x}$ is $\{\bar{x} \in \mathbb{R} : 33.5 < \bar{x} < 34.1\}$. Remarks: $\bar{X} \sim N\left(\mu_0, \frac{\sigma^2}{n}\right)$ exactly and not approximately  The figure shows a normal distribution curve for $\bar{X} \sim N\left(33.8, \frac{0.8^2}{30}\right)$. The mean is 33.8. Two points, 33.5 and 34.1, are marked on the x-axis, with the area under the curve between them shaded and labeled as 0.95. The areas in the tails are labeled as 0.025. Below the graph are two screenshots of a calculator's normal distribution function. The first screenshot shows the 'invNorm' function with 'area: 0.95', '$\mu: 33.8$', '$\sigma: 0.8/\sqrt{30}$', and 'Tail: LEFT CENTER RIGHT'. The second screenshot shows the result of the calculation: 'invNorm(0.95, 33.8, 0.8/\sqrt{30})' resulting in the interval '(33.51372894, 34.08627106)'.

8 In this question you should state clearly the values of the parameters of any normal distribution you use.

The masses in grams of apples have the distribution $N(80, 3^2)$ and the masses in grams of oranges have the distribution $N(100, 5^2)$.

- (i) Find the probability that the total mass of 2 randomly selected oranges is less than 205 grams. [2]
- (ii) Find the probability that the total mass of 4 randomly selected apples is more than three times the mass of 1 randomly selected orange. [3]

During the Community Health Week Event, visitors who take part in the activities are given gift boxes. Each gift box contains 3 apples and 2 oranges that are individually machine wrapped. The mass of wrapper used for each fruit is dependent on the mass of the fruit resulting in the mass of each wrapped fruit being 7% more than the mass of the fruit. The fruits are packed in a gift box and the mass of an empty gift box is normally distributed with mean 50 grams and standard deviation 2 grams. During packing, parts of the gift box is cut and removed. This process reduces the mass of each empty gift box by 10%.

- (iii) The probability that the total mass of a randomly selected gift box is less than k grams is 0.8. Find the value of k .

Find the probability that, at a particular collection point, the 25th gift box that is given is the 19th gift box whose total mass is less than k grams. [5]

(i) [2]	Let X be the mass of an orange in grams. $X \sim N(100, 5^2)$ $X_1 + X_2 \sim N(2 \times 100, 2 \times 5^2)$ $X_1 + X_2 \sim N(200, 50)$ $P(X_1 + X_2 < 205) = 0.760$ (3 s.f.)	Remember to write down the distributions highlighted in blue.
(ii) [3]	Let Y be the mass of an apple in grams. $Y \sim N(80, 3^2)$ $P(Y_1 + Y_2 + Y_3 + Y_4 > 3X) = P(Y_1 + Y_2 + Y_3 + Y_4 - 3X > 0)$ Let $W = Y_1 + Y_2 + Y_3 + Y_4 - 3X$ $E(W) = 4E(Y) - 3E(X) = 4 \times 80 - 3 \times 100 = 20$ $\text{Var}(W) = 4\text{Var}(Y) + 3^2\text{Var}(X) = 4 \times 3^2 + 9 \times 5^2 = 261$ $W \sim N(20, 261)$ $P(W > 0) = 0.892$ (3 s.f.)	
(iii) [5]	Let V be the mass of an empty gift box. $V \sim N(50, 2^2)$ Let M be the total mass of a gift box. $M = 1.07(Y_1 + Y_2 + Y_3 + X_1 + X_2) + 0.9V$ $E(M) = 1.07[3E(Y) + 2E(X)] + 0.9E(V)$ $= 1.07(3 \times 80 + 2 \times 100) + 0.9 \times 50 = 515.8$ $\text{Var}(M) = 1.07^2[3\text{Var}(Y) + 2\text{Var}(X)] + 0.9^2\text{Var}(V)$ $= 1.07^2(3 \times 3^2 + 2 \times 5^2) + 0.9^2 \times 2^2 = 91.3973$ $M \sim N(515.8, 91.3973)$ $P(M < k) = 0.8$ Using GC, $P(M < 523.8) = 0.8$ $\therefore k = 524$ (3 s.f.)	Calculate the variance carefully. Common mistake: Forgetting to square the constants “1.07” and “0.9”.

Required probability $= P(\text{of the first 24 gift boxes given, 18 has total mass less than } k \text{ g}) \times$ $P(25^{\text{th}} \text{ gift box has total mass less than } k \text{ g})$ $= \left[\binom{24}{18} (0.8)^{18} (0.2)^6 \right] (0.8) = 0.124 (3 \text{ s.f.})$	
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- 9 The age of a species of trees can be determined by counting its rings, but that requires either cutting a tree down or extracting a sample from the tree's core. A forester attempts to find a relationship between a tree's age and its diameter instead. Based on past records, the forester found data for the diameter and the age (determined by the counting of its rings) of 8 trees of the same species that had been cut down. The results are given in the following table.

Diameter, D (inches)	1.8	6.6	9.9	10.8	12.8	13.2	15.4	16.5
Age, t (years)	5	8	16	12	22	28	30	40

- (i) Draw a scatter diagram for these values, labelling the axes clearly. [2]
- (ii) Find, correct to 4 decimal places, the product moment correlation coefficient between D and t . Comment on whether a linear model would be appropriate. [2]
- (iii) It was suggested that the tree's age can be modelled by one of the formulae

$$t = a + bD^2, \quad t = ae^{bD},$$

where a and b are constants.

Find, correct to 4 decimal places, the product moment correlation coefficient between

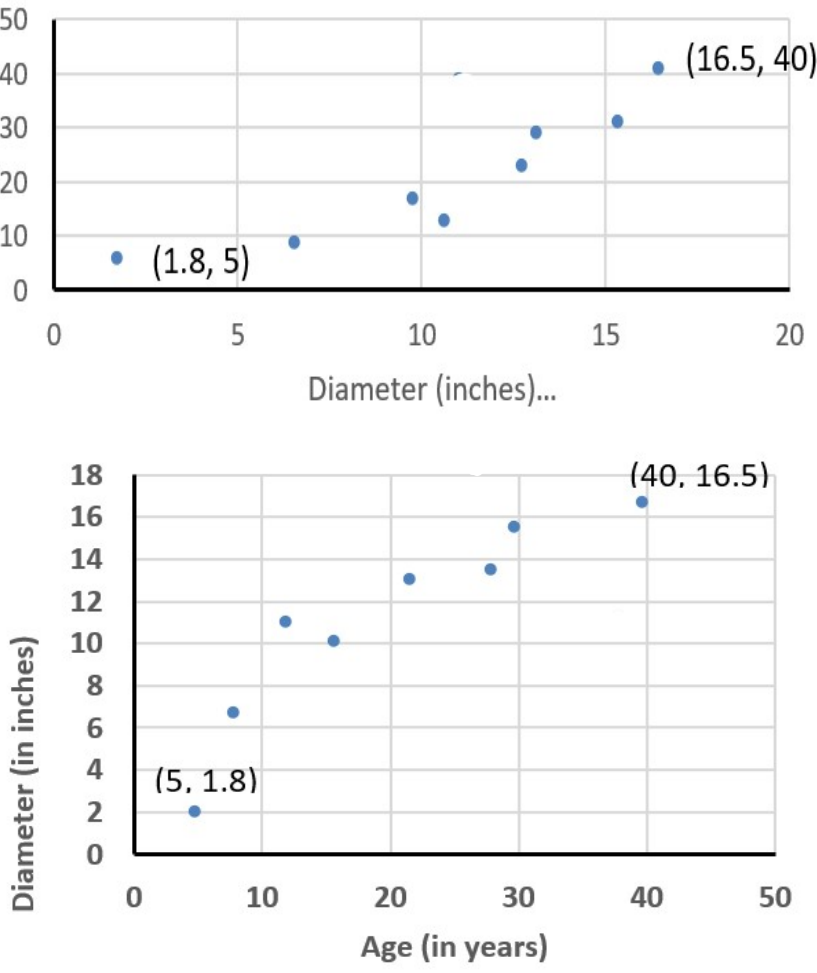
(A) t and D^2 ,

(B) $\ln t$ and D .

Explain which of $t = a + bD^2$ and $t = ae^{bD}$ best models the age of this species of trees based on the given data.

- (iv) The forester wants to estimate the diameter of a tree given its age. Using a suitable regression line, find the required estimate of a tree that is 50 years old. Comment on the reliability of your estimate.

Explain why neither the regression line of t on D^2 nor the regression line of $\ln t$ on D should be used.

(i) [2]	 The figure consists of two scatter plots. The top plot has 'Diameter (inches)' on the y-axis (0 to 50) and 'Age (in years)' on the x-axis (0 to 20). It shows several data points, with two specifically labeled: (1.8, 5) and (16.5, 40). The bottom plot has 'Diameter (in inches)' on the y-axis (0 to 18) and 'Age (in years)' on the x-axis (0 to 50). It shows several data points, with two specifically labeled: (5, 1.8) and (40, 16.5).	Use ruler to draw axes. Label axes, origin. Ensure graph is of reasonable size. Indicate the data range clearly.
(ii) [2]	From GC, the product moment correlation coefficient between D and t is $r = 0.9146$ (4 d.p.) Since r is close to 1 and from the scatter diagram a positive linear correlation between D and t can be seen, a linear model is appropriate. Alternatively Although r is close to 1, scatter diagram suggests that as t increases, D is increasing at a slower rate, so a non-linear model might be more appropriate.	4 decimal places is demanded in the question.
(iii) [2]	From GC, the product moment correlation coefficient between D^2 and t is $r = 0.9689$ (4 d.p.) From GC, the product moment correlation coefficient between D and $\ln t$ is $r = 0.9747$ (4 d.p.) The $t = ae^{bD}$ provides a better model as its product moment correlation coefficient is closer to 1.	4 decimal places is demanded in the question.
(v) [4]	From GC, the regression line of Diameter, D, on $\ln(\text{age})$, $\ln t$, is $D = -7.642653677 + 6.595868978 \ln t$. $t = 50 \Rightarrow \ln t = 3.9120$ $D = 18.161$ (5 s.f.) = 18.2 inches (1 d.p.) The estimated diameter of a 50 year-old tree is 18.2 inches (1 d.p.) As the value of $t = 50$ is outside the data range, the linear model may no longer be suitable. Hence the estimate is not reliable. To estimate D given t, we would need t, the age of the tree, to be the independent variable.	Use at least 5 s.f. for the intermediate working and express final answer in 3 s.f. for non-exact answer (unless specified otherwise).

- (a) Find the number of ways in which Amy can arrange the ten ornaments in a line. [1]

(b) Find the number of ways in which Amy can arrange the ten ornaments in a line such that exactly five heart shaped ornaments are next to each other. [3]

- (c) Amy then decides to arrange the ten ornaments in a circular manner. Find the number of ways in which all ornaments of the same colour are next to each other. [2]

- (d) Amy randomly distributes the ten ornaments into one group of four and two groups of three.

- (i) Show that the probability that the black and white ornaments are in the same group is $\frac{4}{15}$. [3]

- (ii) Find the probability that there are at least two heart shaped ornaments in the group of four given that the black and white ornaments are in the same group. [3]

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	$= \left(\binom{6}{5} 5! \right) 6! - 6! 5! 2$	(12345)6ABCD 1(23456)ABCD are treated as different																
(c) [2]	2 Red, 2 Green, 2 Yellow, 2 Purple, 1 Black and 1 White Number of ways all the ornaments of the same colours are next to each other, arranged in a circular manner $= (6 - 1)! \times 2^4 = 1920$	Remember to arrange all 4 groups.																
(d)(i) [3]	Number of ways to select the three groups $= \frac{\binom{10}{4} \binom{6}{3} \binom{3}{3}}{2!} = 2100$ Number of ways to select when black and white are in the same group = Number of ways when black and white are in group of 4 + number of ways when black and white are in a group of 3 $= \frac{\binom{8}{2} \binom{6}{3} \binom{3}{3}}{2!} + \binom{8}{4} \binom{4}{1} \binom{3}{3} = 560$ Required probability $= \frac{560}{2100} = \frac{4}{15}$ Alternative solutions : $\frac{4}{10} \times \frac{3}{9} + 2 \left(\frac{3}{10} \times \frac{2}{9} \right) \quad \text{or} \quad \frac{\binom{4}{2} + 2 \binom{3}{2}}{\binom{10}{2}} \quad \text{or} \quad \frac{\binom{8}{2}}{\binom{10}{4}} + \frac{2 \binom{8}{1}}{\binom{10}{3}}$	Candidates to know alternative solutions. In a “show question”, more explicit working is required.																
(d)(ii) [3]	Let A be the event that at least 2 heart shaped ornaments are in the group of 4. Let B be the event that the black and white ornaments are in the same group. $P(A B) = 1 - P(A' B) = \frac{n(B) - n(A' \cap B)}{n(B)} \quad \text{or} \quad 1 - \frac{P(A' \cap B)}{P(B)}$ The cases for $A' \cap B$ are as follows: <table><tr><th>Group of 4</th><th>Group of 3</th><th>Group of 3</th><th>Number of Ways</th></tr><tr><td>3 stars with a Red or Green or Yellow or Purple heart</td><td>Black and white and 1 other</td><td>The remaining</td><td>$\binom{4}{3} \binom{4}{1} \binom{4}{1} \binom{3}{3}$ $= 64$</td></tr><tr><td>4 stars</td><td>Black and white and 1 other</td><td>The remaining</td><td>$\binom{4}{4} \binom{4}{1} \binom{3}{3} = 4$</td></tr><tr><td></td><td></td><td></td><td>Total = 68</td></tr></table>	Group of 4	Group of 3	Group of 3	Number of Ways	3 stars with a Red or Green or Yellow or Purple heart	Black and white and 1 other	The remaining	$\binom{4}{3} \binom{4}{1} \binom{4}{1} \binom{3}{3}$ $= 64$	4 stars	Black and white and 1 other	The remaining	$\binom{4}{4} \binom{4}{1} \binom{3}{3} = 4$				Total = 68	Consider cases carefully
Group of 4	Group of 3	Group of 3	Number of Ways															
3 stars with a Red or Green or Yellow or Purple heart	Black and white and 1 other	The remaining	$\binom{4}{3} \binom{4}{1} \binom{4}{1} \binom{3}{3}$ $= 64$															
4 stars	Black and white and 1 other	The remaining	$\binom{4}{4} \binom{4}{1} \binom{3}{3} = 4$															
			Total = 68															

$$P(A|B) = 1 - \frac{\frac{68}{2100}}{\frac{4}{15}} \quad \text{or} \quad \frac{560 - 68}{560} \left(\frac{492}{560} \right)$$

$$= \frac{123}{140} \quad \text{or} \quad 0.879 \text{ (3s.f, 0.87857(5s.f.))}$$

Alternatively, have to consider 4 cases.

Group of 4	Group of 3	Group of 3	Number of Ways
Black and white and 2 others	Any remaining 3	The remaining	$\binom{8}{2} \binom{6}{3} \binom{3}{3} / 2$ $= 280$
2 hearts (not black and white) + 2 stars	Black and white and 1 other	The remaining	$\binom{4}{2} \binom{4}{2} \binom{4}{1} \binom{3}{3} = 144$
3 hearts (not black and white) + 1 stars			$\binom{4}{3} \binom{4}{1} \binom{4}{1} \binom{3}{3} = 64$
4 hearts (not black and white)			$\binom{4}{4} \binom{4}{1} \binom{3}{3} = 4$
			Total = 492

$$P(A|B) = \frac{492}{560}$$

$$= \frac{123}{140} \quad \text{or} \quad 0.879 \text{ (3s.f, 0.87857(5s.f.))}$$

- 11** The salad bar at a restaurant has n separate bowls containing different ingredients. Each customer to the restaurant is allowed only one visit to the salad bar. Jon visits the salad bar and makes a selection. At each bowl, he can take some or none of the contents, and he does not return to the bowl again. As the salad bar may not serve the same ingredients everyday, Jon does not know what ingredients will be served at each visit. On average, the probability that he takes some of the contents from each bowl is 0.4. For each visit to the salad bar, the number of different ingredients Jon takes is the random variable X .

- (i) State, in the context of this question, an assumption needed to model X by a binomial distribution, and explain why the assumption may not hold. [2]

Assume now that X has the distribution $B(n, 0.4)$.

- (ii) Jon visits the salad bar on a randomly chosen day. The probability that Jon takes exactly 7 different ingredients is denoted by P . Determine the value of n that gives the greatest value of P . [2]

Jon's selection on any day is independent of his selection on any other day. During a promotional week, the salad bar serves 6 different ingredients (A, B, C, D, E, F) on Monday and Tuesday, 6 different ingredients (G, H, I, J, K, L) on Wednesday and Thursday, and all 12 different ingredients (A to L) on Friday, Saturday and Sunday.

- (iii) If Jon visits the salad bar on Monday and Thursday of the promotional week, find the probability that he takes at least 3 different ingredients on each day. [2]
- (iv) If Jon visits the salad bar on Saturday of the promotional week, find the probability that he takes at least 6 different ingredients. [1]
- (v) Explain in context why the answer to part (iv) is greater than the answer to part (iii). [1]

After the promotional week, the restaurant decides to charge customers \$3 for a visit to the salad bar, and an additional \$2 if they take more than $\frac{n}{2}$ different ingredients. Kai visits the salad bar after the promotional week and makes a selection. The number of different ingredients Kai takes is the random variable K . It is given that

$$P(K \leq k) = \left(\frac{k}{n}\right)^2, \quad k = 0, 1, 2, \dots, n.$$

- (vi) If Kai visits the salad bar on a particular day after the promotional week when $n = 9$, find the expected amount Kai pays for that particular visit. [3]

Kai visits the salad bar on a randomly chosen day after the promotional week.

- (vii) Show that $P(K = k) = \frac{2k-1}{n^2}$, $k = 1, 2, \dots, n$. [1]

- (viii) Find, in terms of n , the expected number of different ingredients Kai takes on that day.

[You may use the result $\sum_{r=1}^n r^2 = \frac{n}{6}(n+1)(2n+1)$.] [2]

(i) [2]	The probability of Jon taking the ingredients from a bowl remains constant at 0.4 for all bowls. However, the probability may not be constant as Jon may prefer certain ingredients and be more likely to take the ingredients from some bowls than others. OR The event that Jon takes the ingredients from one bowl is independent of the event that he takes the ingredients from another bowl. However, Jon may not take the ingredients independently as he may consider that some ingredients go well together while others do not.	Correct phrasing in context is required.								
(ii) [2]	$X \sim B(n, 0.4)$ $P = {}^nC_7(0.4)^7(0.6)^{n-7}$ Find n such that P has the greatest value. <table><tr><td>n</td><td>P</td></tr><tr><td>16</td><td>0.18889</td></tr><tr><td>17</td><td>0.19267</td></tr><tr><td>18</td><td>0.18916</td></tr></table>	n	P	16	0.18889	17	0.19267	18	0.18916	
n	P									
16	0.18889									
17	0.19267									
18	0.18916									

	From GC, $n=17$. Alternatively, $P(n) \geq P(n-1)$ ${}^nC_7(0.4)^7(0.6)^{n-7} \geq {}^{n-1}C_7(0.4)^7(0.6)^{n-1-7}$ $\frac{n!}{7!(n-7)!}0.6 \geq \frac{(n-1)!}{7!(n-1-7)!}$ $\left(\frac{n}{n-7}\right)0.6 \geq 1$ $0.6n \geq n-7$ $n \leq \frac{35}{2}$ P has greatest value when $n = 17$
(iii) [2]	When $n = 6$, $X \sim B(6, 0.4)$. P(at least 3 different ingredients each on Mon and Thur) $= [P(X \geq 3)]^2 = [1 - P(X \leq 2)]^2 = 0.208 \text{ (3 s.f.)}$
(iv) [1]	When $n = 12$, $X \sim B(12, 0.4)$. P(at least 6 different ingredients on Sat) = $P(X \geq 6) = 1 - P(X \leq 5) = 0.335 \text{ (3 s.f.)}$
(v) [1]	Part (iv) includes the cases in part (iii) as well as other cases. For example, the case of Jon taking exactly 2 different ingredients from A to F and exactly 4 different ingredients from G to L is included in part (iv) but not in part (iii).
(vi) [3]	$P(K > 4.5) = P(K \geq 5) = 1 - P(K \leq 4) = 1 - \left(\frac{4}{9}\right)^2 = 1 - \frac{16}{81} = \frac{65}{81}$ Expected amount Kai pays = $\\$ \left[3 + 2 \left(\frac{65}{81} \right) \right] = \\$4.60 \text{ (to nearest cent)}$ OR alternatively, $\\$ \left[3 \left(\frac{16}{81} \right) + 5 \left(\frac{65}{81} \right) \right] = \\$4.60 \text{ (to nearest cent)}$
(vii) [1]	$P(K = k) = P(K \leq k) - P(K \leq k-1), \quad k = 1, 2, \dots, n$ $= \left(\frac{k}{n}\right)^2 - \left(\frac{k-1}{n}\right)^2$ $= \frac{k^2 - (k^2 - 2k + 1)}{n^2}$ $= \frac{2k-1}{n^2} \text{ (shown)}$
(viii) [2]	Expected number of different ingredients Kai takes $= \sum_{k=0}^n k P(K = k)$ $= \frac{1}{n^2} \sum_{k=1}^n (2k^2 - k)$ $= \frac{1}{n^2} \left[\frac{2n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} \right]$ $= \frac{(n+1)(4n-1)}{6n}$